

Astrophysics Final Exam

Reconstructed question paper

Name: _____ **Date:** _____

Compiled from the questions recalled in the chat. Wording and missing choices have been reconstructed; this is not an official exam copy. Questions 24–26 were not recalled. No answer key is included.

Part I: Multiple choice (Questions 1–26)

Select the best answer for each recalled question.

1. In the lab frame, a strong shock moves through an initially stationary medium with velocity U . What is the velocity of the downstream gas in the lab frame? Assume a monoatomic ideal gas.

- A. U
- B. $U/4$
- C. $3U/4$
- D. $U/2$

2. The supernova remnant SN 1006 is observed in X-rays. What is the most likely origin of the X-ray emission shown in the original exam image? (*Original image not supplied.*)

- A. Thermal blackbody radiation from hot gas
- B. Synchrotron radiation from relativistic electrons
- C. Bremsstrahlung emission from non-relativistic electrons
- D. Line emission from heavy elements only

3. Which of the following is a continuous energy-loss process for high-energy particles?

- A. Nuclear interaction with matter
- B. Spallation
- C. Photopion production
- D. Ionization losses

4. For a strong shock in a monoatomic gas ($\gamma = 5/3$), what is the compression ratio r ?

- A. 1
- B. 2
- C. 3
- D. 4

5. What is a major advantage of space-based cosmic-ray experiments compared with ground-based experiments?
- A. Better angular resolution
 - B. Better energy resolution
 - C. Larger effective detector area
 - D. Better charge (atomic-number) resolution
6. In AMS-02 and Fermi-LAT, which detector component is used for calorimetric energy measurements?
- A. Silicon tracker
 - B. Time-of-flight system
 - C. Calorimeter
 - D. Cherenkov detector
7. Electrons in a source have a power-law spectrum $N_e(E) \propto E^{-2}$. For bremsstrahlung emission from these electrons, what is the approximate exponent of the differential photon spectrum?
- A. -1
 - B. -2
 - C. -3
 - D. -4
8. Why is nitrogen considered both a primary and a secondary cosmic-ray nucleus?
- A. It is produced only in supernova ejecta.
 - B. It is accelerated at sources and is also produced by spallation of heavier nuclei in the interstellar medium.
 - C. It is produced only by spallation of hydrogen and helium.
 - D. It is produced exclusively in the solar wind.
9. Which observational quantity most directly provides a radioactive clock for estimating the confinement time of cosmic rays in the Galaxy?
- A. Ratio of primary to secondary nuclei
 - B. Ratio of unstable to stable secondary nuclei
 - C. Shape of the secondary spectrum
 - D. All of the above

- 10.** The energy dependence of the ratio of secondary to primary cosmic-ray nuclei is mainly determined by:
- A. The source spectral index only
 - B. The energy dependence of the diffusion coefficient
 - C. A ratio independent of both the source spectrum and energy
 - D. A ratio independent of energy but dependent on charge
- 11.** In diffusive shock acceleration (first-order Fermi acceleration), the fractional energy gain per cycle scales with shock speed u approximately as:
- A. u/c
 - B. $(u/c)^2$
 - C. c/u
 - D. $(c/u)^2$
- 12.** Suppose the primary source spectrum is $Q(E) \propto E^{-\alpha}$ and the diffusion coefficient is $D(E) \propto E^a$. In a steady state dominated by diffusive escape, what are the primary and secondary spectra?
- A. Primaries: $E^{-\alpha}$; secondaries: $E^{-\alpha}$
 - B. Primaries: $E^{-(\alpha+a)}$; secondaries: $E^{-(\alpha+a)}$
 - C. Primaries: $E^{-(\alpha+a)}$; secondaries: $E^{-(\alpha+2a)}$
 - D. Primaries: $E^{-(\alpha+2a)}$; secondaries: $E^{-(\alpha+3a)}$
- 13.** The injected electron spectrum is $Q(E) \propto E^{-\lambda}$. At high energies, radiative losses dominate with $b(E) \propto E^2$. What is the steady-state propagated electron spectrum?
- A. $N(E) \propto E^{-\lambda}$
 - B. $N(E) \propto E^{-(\lambda+1)}$
 - C. $N(E) \propto E^{-(\lambda-1)}$
 - D. $N(E) \propto E^{-(\lambda+1/2)}$
- 14.** Which statement about neutron stars is incorrect?
- A. They have very strong magnetic fields.
 - B. They have extremely high densities.
 - C. Their emission beams always point toward Earth.
 - D. They can form in the core collapse of massive stars.

15. Which statement about synchrotron radiation from a relativistic electron is correct? Let γ denote the electron's Lorentz factor and L the orbit length.

- A. Radiation is beamed within an angle of order $1/\gamma$ around the electron's velocity.
- B. Radiation is beamed within an angle of order $1/\gamma^2$.
- C. The observed pulse duration from one orbit is of order L/c .
- D. The observed pulse duration is much longer than L/c .

16. Scattering of charged particles by magnetic turbulence is resonantly enhanced when the particle gyroradius is:

- A. Much larger than the characteristic scale of the magnetic fluctuations
- B. Much smaller than the characteristic scale of the magnetic fluctuations
- C. Comparable to the characteristic scale of the magnetic fluctuations
- D. Exactly half the characteristic scale of the magnetic fluctuations

17. In high-energy proton–proton collisions producing pions, $pp \rightarrow \pi^0, \pi^\pm + \dots$, approximately how much of a proton's initial kinetic energy E is carried by a final electron or positron from a charged-pion decay chain?

- A. $E/2$
- B. $E/6$
- C. $E/12$
- D. $E/18$

18. A possible source contribution invoked to explain the helium abundance and spectrum in cosmic rays is:

- A. Wolf–Rayet stars
- B. White dwarfs in binary systems
- C. Planets in the solar system
- D. Brown dwarfs

19. Which is a key prediction of diffusive shock acceleration for cosmic-ray nuclei?

- A. The spectral index depends strongly on nuclear mass number.
- B. The spectral index depends strongly on nuclear charge Z .
- C. The spectral index is approximately independent of the type of nucleus.
- D. Only protons can be accelerated efficiently.

20. What is the major energy-loss mechanism for heavy cosmic rays at moderate energies in the interstellar medium?

- A. Ionization losses
- B. Spallation
- C. Interaction with magnetic fields

21. In ground-based air-shower experiments, the depth of shower maximum $X_{\max}$ (or height $h_{\max}$) is primarily used to determine:

- A. The type (mass) of the primary particle
- B. The energy of the primary particle
- C. The charge of the primary particle
- D. All of the above equally

22. The maximum energy at which a cosmic-ray particle can be confined in a given accelerator is primarily limited by:

- A. The mass of the particle
- B. The charge of the particle
- C. The speed of the particle
- D. The gyroradius of the particle relative to the size of the accelerator

23. The interstellar medium contains:

- A. Magnetic fields only
- B. Matter (gas and dust) only
- C. Photons only
- D. Matter, magnetic fields, and photons

24. *Question not recalled.*

25. *Question not recalled.*

26. *Question not recalled.*

Part II: Short answer (Questions 27–32)

Show your reasoning and relevant equations.

27. In the diffusive model of Galactic cosmic-ray propagation, why must nearby sources sometimes be treated using a discrete space–time description rather than a smooth, continuous source distribution? Discuss the implications for local spectra or secondary-to-primary ratios.

28. Observations show an unexpected excess of positrons above about 10 GeV compared with simple secondary-production models. Give one possible astrophysical or particle-physics explanation.

29. Explain why a proton needs a kinetic energy of approximately 282 MeV to produce a neutral pion with rest energy $m_{\pi^0}c^2 \simeq 135$ MeV in the reaction

$$p + p \longrightarrow p + p + \pi^0,$$

with one proton initially at rest. Outline the kinematic threshold argument.

Part II: Short answer (continued)

30. Neutrinos interact only weakly with matter, yet large water or ice Cherenkov detectors can detect them. Explain how light sensors (photomultiplier tubes or optical modules) detect neutrino events in these detectors.

31. State the approximate rest-frame lifetime of the muon. Briefly explain how time dilation affects the observation of atmospheric muons at ground level.

32. Consider a steady-state balance equation for electrons with an energy-loss rate $b(E) \propto E$ and source term $Q(E) \propto E^\lambda$:

$$\frac{d}{dE} [b(E)N(E)] + Q(E) = 0.$$

Derive the steady-state spectrum $N(E)$ up to a constant factor and identify its power-law index. State the boundary condition and any restriction on λ needed for your solution.
